Premature Accountability Convergence in Regulated Organizations

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Abstract

Responsibility in regulated organizations does not emerge only after failure — it is continuously produced across anticipatory alignment, procedural embedding, and retrospective stabilization. As artificial intelligence systems become integrated into documentation and evaluative infrastructures, responsibility is increasingly mediated through computational representation. Yet AI does not create new accountability structures. Instead, it reshapes the temporal dynamics through which responsibility becomes stabilized.

This paper introduces **Premature Accountability Convergence (PAC)** to describe a conditional acceleration dynamic within the responsibility lifecycle. PAC occurs when computational mediation compresses interpretive space such that responsibility stabilizes more rapidly than causal ambiguity can be meaningfully examined. Under conditions of representational compression, epistemic authority amplification, and temporal contraction of sensemaking, responsibility may converge before alternative structural interpretations remain viable.

The analysis is grounded in two empirical components. First, document analysis of FDA Warning Letters ($N = 50$, 2016–2025, finished pharmaceutical manufacturing) establishes that responsibility convergence is a structural property of accountability documentation in regulated environments prior to any computational

mediation: across 160 violation observations, attribution language varies substantially — action-oriented ("failure to," 24.4%) versus state-oriented ("inadequate," 66.2%) — yet final accountability converges uniformly onto the regulated firm in 159 of 160 cases. Documentary form compresses interpretive diversity into administratively actionable attribution regardless of causal framing. Second, a two-study empirical program in pharmaceutical investigation contexts demonstrates both the behavioral baseline and its design implications. A pilot study ($N = 20$) shows that under closure pressure, 15 of 20 participants produced single-cause explanations at the initial timepoint, with confidence ratings averaging 1 out of 5 regardless of experience level — indicating that early convergence reflects institutional pressure rather than epistemic certainty. A scaffolded AI intervention ($N = 20$) demonstrated that counter-convergence design can delay premature closure — 18 of 20 participants broadened causal reasoning, 15 of 20 reconsidered initial attribution, and 12 of those 15 shifted from person-centered to systemic or procedural explanations, with usability ratings averaging 5.0 out of 5 and all 20 expressing adoption intent — though replication at scale remains necessary.

By conceptualizing AI as a representational accelerator rather than an autonomous decision-maker, this framework shifts attention from algorithmic bias toward the temporal architecture of accountability. PAC extends sociotechnical theories of responsibility by specifying how computational infrastructures intensify stabilization dynamics already embedded within organizational systems — and by situating this acceleration within a broader multi-stage responsibility lifecycle spanning anticipatory alignment (AAC), procedural embedding (CF), investigative stabilization (RC), and AI-mediated representational acceleration (PAC).

Keywords: premature accountability convergence, artificial intelligence, responsibility lifecycle, representational compression, organizational investigations, regulated environments, pharmaceutical manufacturing

1 Introduction

In 159 of 160 FDA Warning Letter violations issued to pharmaceutical manufacturers between 2016 and 2025, final accountability converges onto the regulated firm — regardless of whether the violation is described as an actor's "failure to" act or an "inadequate" organizational system. This pattern is not a product of computational mediation. It reflects a more fundamental dynamic: in regulated organizations, documentary form has long compressed interpretive space and concentrated accountability onto identifiable actors before any AI system enters the sensemaking process.

Yet as artificial intelligence systems become embedded within documentation, evaluation, and compliance infrastructures, this pre-existing convergence dynamic does not disappear — it accelerates. AI systems generate structured representations — summaries, classifications, anomaly flags, predictive indicators — that enter directly into accountability workflows. While frequently described as decision-support tools, these systems operate at a deeper infrastructural level: they intervene in the representational architecture through which responsibility becomes legible, assessable, and administratively actionable.

This pattern establishes a critical baseline: responsibility convergence is a structural property of accountability documentation in regulated environments, operating independently of computational mediation. What AI systems introduce is not convergence itself, but acceleration — a reshaping of the temporal dynamics through which interpretive space narrows and responsibility stabilizes.

Responsibility in regulated organizations is not a singular event but a temporally structured accomplishment. It is anticipatorily aligned before action, procedurally embedded during execution, and retrospectively stabilized under conditions of ambiguity and closure pressure. Prior work within this responsibility program has conceptualized how expectations converge prospectively (Anticipated Accountability Convergence, AAC) and how attribution stabilizes retrospectively through investigative reasoning (Responsibility Convergence, RC). These frameworks explain how responsibility becomes structured and consolidated across stages of organizational life.

What remains theoretically underspecified, however, is the temporal architecture of

convergence itself. Existing theories clarify how responsibility aligns and stabilizes; they do not fully account for how infrastructural mediation may alter the tempo and breadth of that stabilization. As AI systems become embedded within documentation and evaluative processes, interpretive space may narrow earlier in the sensemaking sequence. Convergence may occur more rapidly — not because actors intend closure, nor because institutions become more rigid, but because computational mediation compresses representational diversity and accelerates inscription.

Without a framework that isolates this acceleration dynamic, the effects of AI on accountability risk being misattributed to bias, individual cognition, or institutional design alone. The missing dimension is temporal: how quickly responsibility stabilizes, how much interpretive variance persists before closure, and how infrastructural mediation reshapes these dynamics.

This paper introduces **Premature Accountability Convergence (PAC)** to conceptualize this acceleration mechanism. PAC identifies the conditions under which computational mediation compresses interpretive space and hastens responsibility stabilization within the responsibility lifecycle. Rather than asking whether AI changes who is responsible, this framework asks how AI reshapes the timing, durability, and scope of responsibility convergence across anticipatory alignment, procedural embedding, and retrospective stabilization. In doing so, it advances a temporal turn in the theory of organizational accountability.

2 Responsibility as a Multi-Stage Organizational Lifecycle

Responsibility in regulated sociotechnical organizations unfolds across temporally distinct yet structurally interconnected stages. Rather than emerging solely at the moment of investigation or formal evaluation, responsibility is progressively produced through anticipatory alignment, procedural embedding, and retrospective stabilization (Perrow, 1984; Vaughan, 1996). These stages together constitute a lifecycle through which accountability

expectations are generated, operationalized, and ultimately consolidated into administratively actionable forms.

Crucially, responsibility convergence does not occur at a single point. It accumulates across stages. Interpretive variance narrows prospectively, is structured evidentially during execution, and is formally condensed under retrospective closure pressure. Understanding this lifecycle clarifies how accountability is temporally organized prior to the introduction of computational mediation.

Anticipatory Structuring of Responsibility

In regulated environments, organizational actors rarely act without anticipation of future evaluation. Accountability expectations operate prospectively, shaping how obligations, risks, and acceptable conduct are interpreted before execution. The possibility of inspection, audit, or review functions as a structuring condition of action rather than a subsequent add-on.

Anticipatory alignment occurs when actors orient behavior toward expected evaluative standards. Responsibility is therefore structured ex ante: individuals interpret instructions, exercise discretion, and assess risk with awareness of how these actions may later be scrutinized. Such alignment does not require explicit sanction at the moment of execution; rather, it emerges from embedded expectations that future justification will be required. In pharmaceutical GMP environments, for instance, quality control analysts document interpretive decisions not merely to record outcomes but to produce accounts that will withstand regulatory inspection — orienting behavior toward anticipated auditability before any deviation or investigation occurs (Power, 1997).

At this stage, convergence operates prospectively. Interpretive variance narrows not through formal attribution, but through alignment with anticipated evaluative criteria. Responsibility functions as a precondition for action, organizing conduct around the expectation of later assessability.

Procedural Embedding During Execution

During execution, responsibility expectations become embedded within documentation systems, reporting architectures, and procedural infrastructures. Standard operating procedures, logging requirements, validation protocols, and compliance frameworks structure how events are recorded and translated into organizational memory.

Procedural embedding stabilizes responsibility not by assigning blame, but by structuring the evidentiary pathways through which future attribution will unfold. Documentation formats define what counts as relevant information, how deviations are categorized, and which explanatory variables become visible within formal records. In doing so, procedural infrastructures pre-format the representational space within which retrospective interpretation will occur. The cognitive friction embedded within procedural language — conditional branching, cross-references, exception structures, and discretionary qualifiers — further structures which judgment demands become individually attributable during execution.

Convergence at this stage remains incomplete but accumulative. Interpretive possibilities are not eliminated; rather, they are channeled into administratively legible categories. Responsibility is progressively shaped through the structuring of evidence, even before any explicit investigative closure takes place.

Retrospective Stabilization of Responsibility

After outcomes occur, responsibility is articulated and consolidated through investigations, reports, evaluative documentation, and corrective action processes. Under conditions of causal ambiguity, organizations must translate distributed events into administratively actionable accounts. This requires selecting attributional anchors that can stabilize accountability within formal structures.

Investigative reasoning frequently unfolds under closure pressure: ambiguous causal pathways must be rendered determinate in order to justify corrective action, compliance responses, or performance evaluation (Dekker, 2007; Vaughan, 1996). Through this process, interpretive alternatives are gradually reduced as coherent explanatory narratives

acquire institutional authority. The FDA Warning Letter data illustrates this dynamic at the regulatory level: across violation categories ranging from investigation failures to data integrity lapses, final attribution converges onto the regulated firm regardless of whether the evidentiary basis supports actor-centered or system-centered causal accounts.

Retrospective stabilization marks the formal consolidation of responsibility convergence. Ambiguity is condensed into administratively durable accounts that become embedded in documentation systems and organizational memory. At this stage, responsibility is no longer anticipatory but stabilized; convergence is driven by the need for closure and institutional legibility (Weick, 1995).

Taken together, these stages illustrate that responsibility convergence is not a discrete event but a cumulative and temporally distributed organizational process. Each stage narrows interpretive variance, structures evidentiary pathways, and consolidates accountability in ways that shape the conditions under which subsequent stages unfold. This lifecycle provides the structural foundation against which computational mediation — and its acceleration of convergence dynamics — must be understood.

3 Computational Mediation within the Responsibility Lifecycle

Having established responsibility as a temporally structured organizational lifecycle, we now consider how computational systems enter and interact with this process. Artificial intelligence systems are increasingly embedded within procedural workflows, documentation regimes, and evaluative infrastructures across regulated domains. Rather than functioning as autonomous decision-makers, these systems operate as representational mediators that translate organizational events into structured, recordable, and evaluable forms.

Computational outputs — such as summaries, classifications, anomaly flags, pattern detections, and predictive indicators — do not independently assign responsibility. However, they shape how events are rendered legible within organizational documentation

and evaluative processes. By structuring information into standardized and repeatable representations, computational mediation influences how actors interpret, record, and retrospectively reconstruct events. In pharmaceutical quality systems, for instance, electronic quality management platforms such as Veeva Vault structure deviation investigation workflows through mandatory completion fields, approval routing, and classification schemas — converting open-ended causal inquiry into sequenced administrative processes with predefined closure criteria. The representational architecture of these systems does not determine investigative conclusions, but it shapes which causal pathways remain visible and administratively actionable within formal documentation.

Importantly, computational mediation does not introduce responsibility structures ex nihilo. The anticipatory alignment, procedural embedding, and retrospective stabilization of responsibility precede the integration of AI systems. As the FDA Warning Letter analysis demonstrates, organizational infrastructures already generate expectations of accountability and stabilize attribution through documentary form — independently of computational mediation. Across 160 violation observations, the representational architecture of regulatory documentation compresses interpretive diversity into administratively actionable attribution regardless of causal framing. What computational mediation alters is not the existence of responsibility convergence but its tempo and representational form.

Because algorithmically generated representations often carry the perceived authority of formalized calculation, they may narrow interpretive divergence earlier in the sensemaking process. Structured outputs can foreground particular causal pathways while backgrounding others, shaping which interpretations appear administratively actionable. In doing so, computational systems may compress the temporal space between ambiguity and stabilization within the responsibility lifecycle.

This interaction introduces a distinct dynamic: responsibility expectations may align and stabilize more rapidly when mediated through computationally structured representations. The lifecycle remains intact, but its stabilization phase may be accelerated through representational compression. Understanding this acceleration dynamic requires

a framework that specifies the conditions under which computational mediation intensifies convergence — and the mechanisms through which it does so.

4 Premature Accountability Convergence (PAC)

Computational mediation does not introduce responsibility structures into organizational systems; rather, it interacts with a pre-existing lifecycle of convergence. Within this lifecycle — anticipatory structuring, procedural embedding, and retrospective stabilization — AI systems function as representational accelerators. They do not determine who is responsible, but they may alter the tempo and breadth through which responsibility stabilizes.

Premature Accountability Convergence (PAC) identifies a structural acceleration tendency that becomes visible under conditions of computational mediation. PAC occurs when representational compression, perceived epistemic authority, and temporal contraction combine to reduce interpretive variance before causal ambiguity has been meaningfully examined. The lifecycle itself remains intact; what shifts is the timing and scope of stabilization.

PAC is not a claim that AI inherently distorts accountability. Rather, it specifies a conditional infrastructural dynamic: when structured computational outputs enter documentation and evaluative systems early in the sensemaking sequence, responsibility may converge more rapidly and with diminished interpretive plurality. Convergence, in such cases, is not imposed but accelerated.

PAC therefore reframes computational mediation as a temporal intervention within accountability architecture. It draws attention to how infrastructural design shapes the durability of ambiguity, the persistence of alternative explanations, and the pace at which attribution becomes administratively actionable.

Three interrelated mechanisms contribute to PAC: representational compression, epistemic authority amplification, and temporal contraction of sensemaking.

Representational Compression

Computational systems translate complex and distributed organizational events into structured outputs — summaries, classifications, risk scores, anomaly flags, and predictive indicators. In doing so, they compress interpretive multiplicity into standardized representational forms.

This compression enhances administrative clarity and traceability. However, it also pre-selects which explanatory variables remain visible within formal documentation. As events are formatted into predefined categories, the range of plausible causal accounts may narrow prior to full deliberation. Interpretive variance does not disappear, but it becomes institutionally backgrounded.

The FDA Warning Letter corpus illustrates how representational compression operates even prior to computational mediation. Across six violation clusters — investigation failures, oversight absence, validation deficiencies, data integrity lapses, release and stability issues, and equipment and environmental conditions — attribution language varies substantially in its causal framing. Yet the documentary form of the Warning Letter compresses this variation into a uniform attributional outcome: firm-level accountability in 159 of 160 cases. The representational architecture of regulatory documentation pre-selects the attributional anchor before any investigative deliberation occurs. Computational systems intensify this dynamic by further standardizing the categories through which organizational events are rendered legible — accelerating the compression that documentary form already initiates.

Representational compression thus operates as the primary structural driver of PAC.

Epistemic Authority Amplification

Algorithmically generated outputs often carry perceived legitimacy derived from formalized calculation, technical infrastructure, and data-driven framing. When computational representations are treated as epistemically authoritative, interpretive contestation may diminish.

Epistemic authority amplification does not require blind trust in algorithms. Rather,

it emerges when structured outputs become embedded within formal documentation systems and are granted institutional weight. Under conditions of ambiguity, such outputs may anchor explanatory frames earlier in the stabilization process, accelerating convergence around computationally foregrounded interpretations.

In regulated environments, this dynamic is observable when electronic quality management systems generate automated deviation classifications or risk scores that become the starting point for investigative reasoning rather than one input among many. When a system flags an event as a procedural non-conformance, investigators may orient their causal reconstruction around that classification — not because they defer uncritically to the algorithm, but because the classification carries institutional authority within the documentation architecture. Alternative systemic or structural explanations may receive diminished scrutiny as a result, even when evidentiary ambiguity would otherwise support their consideration.

Temporal Contraction of Sensemaking

Computational mediation frequently reduces the interval between event occurrence and formal inscription. Automated logging systems, real-time dashboards, and immediate classification tools shorten the deliberative window within which alternative interpretations may circulate informally.

When documentation is generated rapidly and embedded early within evaluative infrastructures, preliminary interpretations acquire durability before distributed explanations are fully articulated. Responsibility may therefore stabilize prior to the exhaustion of interpretive diversity.

Temporal contraction amplifies the acceleration dynamic initiated by representational compression. In pharmaceutical GMP environments, this dynamic is observable in deviation management systems that impose mandatory investigation closure timelines — typically 30 days for minor deviations, with escalation protocols for extensions. These timeframe requirements convert open-ended causal inquiry into time-bounded administrative tasks. When AI-assisted classification tools generate preliminary root cause

categories at the point of deviation initiation, those categories acquire durability within the investigation record before the full evidentiary picture has emerged. The deliberative interval within which alternative interpretations might circulate — across quality, operations, and engineering functions — contracts around computationally inscribed preliminary accounts.

Representational compression constitutes the primary structural driver of PAC, translating distributed organizational events into administratively legible forms. Yet acceleration does not operate uniformly. Its intensity depends on contextual amplification mechanisms — most notably the perceived epistemic authority granted to computational outputs and the temporal interval available for interpretive deliberation. When these conditions converge, responsibility stabilization may occur earlier and with diminished interpretive plurality.

5 Theoretical Propositions

The convergence model yields four theoretical propositions specifying how variation in representational compression, epistemic authority, and temporal contraction conditions the strength and distribution of PAC across organizational contexts.

Proposition 1: Representational Compression

The greater the degree of computational representational compression within organizational documentation systems, the more likely interpretive space will narrow prior to full causal exploration.

Computational representational compression refers to the extent to which complex, distributed, and ambiguous events are translated into standardized, categorical, or summarized formats. As organizational documentation systems rely more heavily on structured computational outputs — such as classification schemas, risk categories, or summary dashboards — interpretive possibilities become formatted into predefined representational containers. While such structuring enhances administrative clarity and efficiency,

it simultaneously limits the range of causal alternatives that remain visible within formal documentation. When ambiguity is compressed into categorical outputs, alternative explanatory pathways may lose salience before being fully explored, thereby increasing the likelihood of premature convergence.

This proposition is non-obvious in an important respect: representational compression does not require intentional simplification or algorithmic bias. It operates structurally — through the design of documentation formats, classification schemas, and reporting architectures — regardless of the intentions of system designers or investigators. The FDA Warning Letter corpus illustrates this at the pre-computational level: the documentary form of regulatory enforcement produces uniform firm-level attribution across causally heterogeneous violations, not because investigators intend to simplify, but because the representational architecture of the document is organized to produce administratively actionable attribution. Computational systems intensify this structural tendency by extending standardized categorization earlier and more densely into the sensemaking sequence.

Proposition 2: Epistemic Authority

The stronger the perceived epistemic authority of computational outputs, the more rapidly responsibility expectations will align around computationally foregrounded interpretations.

Epistemic authority amplification occurs when algorithmically generated outputs are treated as objective, neutral, or technically superior forms of knowledge. Because computational representations derive legitimacy from formalized calculation and technical infrastructure, actors may defer to their outputs with reduced contestation (Porter, 1995). In contexts of ambiguity, perceived data-driven precision can accelerate interpretive alignment by stabilizing particular explanatory frames early in the sensemaking process. As actors converge around computationally foregrounded interpretations, alternative causal accounts may receive diminished scrutiny, thereby contributing to accelerated stabilization within the responsibility lifecycle.

What makes this proposition theoretically significant is the asymmetry it introduces:

epistemic authority amplification does not require that computational outputs be accurate or complete. It requires only that they be institutionally credible. In regulated environments where documentation systems carry audit authority, computationally generated classifications may acquire institutional weight independent of their evidentiary completeness — anchoring investigative reasoning around foregrounded interpretations before the full causal picture has emerged.

Proposition 3: Temporal Contraction

The shorter the temporal interval between event occurrence and formal computational documentation, the greater the likelihood of premature responsibility stabilization.

Temporal contraction of sensemaking refers to the reduction in deliberative interval between an event and its formal inscription within documentation systems. Real-time dashboards, automated incident logging, and immediate classification tools compress the period during which interpretive divergence may be explored informally. When documentation is generated rapidly and becomes embedded within evaluative infrastructures, early interpretations gain durability and institutional visibility. As a result, responsibility accounts may solidify before distributed or systemic explanations are meaningfully articulated, increasing the probability of premature convergence.

This proposition connects PAC to broader organizational theories of sensemaking and closure pressure. Temporal contraction does not eliminate interpretive diversity — it forecloses it institutionally by granting durability to preliminary accounts before alternatives are fully examined. In pharmaceutical deviation management systems, mandatory investigation timelines compound this dynamic: when AI-assisted classification tools generate preliminary root cause categories at the point of deviation initiation, those categories acquire documentary durability within closure-pressured investigation cycles, narrowing the deliberative window available for distributed causal reconstruction.

Proposition 4: Counter-Convergence Design

Deliberate design interventions that expand interpretive time, distribute epistemic authority, and preserve representational plurality can delay or mitigate premature accountability convergence.

PAC is not an inevitable consequence of computational mediation. Its intensity depends on infrastructural conditions — the degree of representational compression, the perceived authority of computational outputs, and the temporal interval available for deliberation. These conditions are design variables. Investigative systems that require explicit comparison across multiple causal hypotheses before closure, that treat computational classifications as provisional inputs rather than authoritative determinations, and that preserve temporal buffering between event inscription and formal attribution can reduce the acceleration dynamic identified by PAC.

This proposition is grounded in the scaffolded intervention component of this study. A structured reasoning scaffold ($N = 20$) demonstrated that counter-convergence design can delay premature closure without automating conclusions or undermining compliance expectations, though replication at scale remains necessary. In the unassisted baseline, 15 of 20 participants produced single-cause explanations at the initial timepoint, with confidence ratings averaging 1 out of 5 across all participants regardless of experience level — indicating that early convergence reflects institutional pressure rather than epistemic certainty. With the scaffold, 18 of 20 participants broadened causal reasoning beyond their initial single-cause explanation; 15 of 20 reconsidered their initial attribution, with 12 of those 15 shifting from person-centered to systemic or procedural explanations. Usability ratings averaged 5.0 out of 5 and all 20 participants expressed adoption intent. The primary adoption barrier was conceptual rather than interactional — the interface was perceived as clear, but the notion that investigation reasoning deserves structured support was unfamiliar in GMP contexts.

6 Empirical Illustration: Responsibility Convergence in FDA Regulatory Documentation

The theoretical framework developed above specifies PAC as a conditional acceleration dynamic within the responsibility lifecycle. Before examining how computational mediation intensifies this dynamic, it is necessary to establish its structural baseline: the extent to which responsibility convergence already operates within regulatory documentation prior to any computational intervention. Analysis of FDA Warning Letters issued to finished pharmaceutical manufacturers provides this empirical grounding.

Data and Method

The corpus consists of 50 FDA Warning Letters issued to finished pharmaceutical drug product manufacturers between 2016 and 2025, sampled purposively at five cases per year to ensure temporal coverage across the analysis window. The unit of analysis is the violation block — the individual observation or citation within each Warning Letter — yielding 160 violation observations across the 50-case corpus. Warning Letters were selected from the finished pharmaceutical manufacturing sector because this context combines high regulatory formalization, mandatory documentation requirements, and consequential accountability attribution, making it an analytically clear site for examining how documentary form structures responsibility convergence.

Each violation block was coded along two dimensions using a rule-based computational coding procedure applied consistently across the full corpus. Attribution language was classified into three categories: action-oriented language ("failure to"), which foregrounds identifiable actor conduct and implies direct behavioral accountability; state-oriented language ("inadequate"), which describes systemic or procedural deficiencies and implies organizational or structural accountability; and residual language (incomplete, lack of, and related formulations), which signals absence or deficiency without specifying a clear attributional direction. Violation type was classified into six clusters: investigation failures (21 CFR 211.192), oversight absence (21 CFR 211.22), validation defi-

ciencies (21 CFR 211.100), data integrity lapses, release and stability issues, and equipment and environmental conditions. The complete coding scripts, criteria definitions, and violation-level dataset are publicly available at `https://github.com/sorakim-lab/fda-warning-letters-pac-analysis`, ensuring full reproducibility of the classification procedure.

Participants in both studies were recruited from within the author's pharmaceutical GMP team; all had direct experience with deviation investigation or CAPA processes.

Findings: Attribution Language and Responsibility Convergence

Across the 160 violation observations, attribution language distributes as follows: action-oriented language appears in 39 violations (24.4%), state-oriented language in 106 violations (66.2%), and residual formulations in 15 violations (9.4%). This distribution indicates that the majority of regulatory violations are framed in terms of organizational or systemic states rather than identifiable actor conduct — suggesting that the evidentiary basis for individual attribution is frequently absent or at least not linguistically foregrounded within Warning Letter documentation.

Despite this variation in attributional framing, final accountability converges uniformly onto the regulated firm. Of 160 violation observations, 159 (99.4%) attribute responsibility to the Firm as the primary accountability target. A single violation observation (0.6%) attributes responsibility to the Quality Control Unit as a distinct organizational subunit. No violation observations attribute responsibility to individual actors, distributed systemic conditions, or external factors.

This convergence is structurally independent of attribution language. Violations framed in action-oriented terms ("failure to investigate," "failure to establish") converge onto the Firm at the same rate as violations framed in state-oriented terms ("inadequate quality oversight," "inadequate procedures"). Documentary form — the Warning Letter as a regulatory instrument — compresses causally heterogeneous violation descriptions into a uniform attributional outcome. The representational architecture of the document produces administratively actionable attribution regardless of the causal account

embedded within individual violation blocks.

Cluster-Level Patterns

Examination of attribution language distribution across violation clusters reveals systematic variation in how different violation types are linguistically framed, while final accountability remains uniformly convergent.

Investigation failures (cluster 1, $n = 18$) exhibit the highest concentration of action-oriented language, with 67% of violations framed as "failure to" conduct required investigative activities. This framing foregrounds identifiable procedural conduct — the failure to initiate, complete, or adequately document investigations — making individual or unit-level accountability linguistically visible within the violation description itself.

Oversight absence violations (cluster 2, $n = 24$) and data integrity lapses (cluster 4, $n = 23$) exhibit predominantly state-oriented language, with 79% and 83% of violations respectively framed as "inadequate" systems, controls, or oversight mechanisms. These framings describe organizational states rather than actor conduct, distributing causal implication across systemic conditions while maintaining Firm-level attribution as the accountability outcome.

Validation deficiency violations (cluster 3, $n = 15$) exhibit exclusively state-oriented language, with 100% of violations framed as "inadequate" written procedures or validation protocols. This cluster most clearly illustrates the representational compression dynamic: violations that are entirely framed in systemic terms nonetheless produce individual Firm-level attribution as their accountability outcome.

Equipment and environmental violations (cluster 6, $n = 38$) similarly exhibit high concentrations of state-oriented language (89%), describing facility conditions and equipment states rather than actor conduct, while maintaining uniform Firm-level convergence.

Theoretical Implications

These findings establish three empirical observations that ground the PAC framework. First, responsibility convergence is a structural property of accountability documentation

in regulated environments, operating prior to and independently of computational mediation. The Warning Letter corpus demonstrates that documentary form produces uniform attribution outcomes across causally heterogeneous violation descriptions — compressing interpretive diversity into administratively actionable accountability assignments regardless of evidentiary framing.

Second, representational compression operates at the level of documentary architecture rather than individual investigative reasoning. The convergence pattern observed across 160 violation observations is not attributable to investigator intent, cognitive bias, or individual attribution tendencies. It reflects the structural organization of regulatory documentary form — its design as an instrument for producing sanctionable accountability within a compliance enforcement system.

Third, this baseline dynamic establishes the condition against which computational acceleration must be understood. PAC does not introduce convergence into organizational systems; it intensifies and accelerates a structural tendency already embedded within accountability documentation. Computational systems that generate preliminary classifications, automated risk scores, or structured deviation summaries interact with — and amplify — a pre-existing representational compression dynamic. The tempo and scope of convergence may therefore shift substantially as AI-mediated representations become embedded earlier and more densely within the sensemaking sequence.

7 Discussion

PAC contributes to sociotechnical and organizational theories of accountability by introducing acceleration as a critical analytical dimension within responsibility structuring. Prior research has examined how responsibility is anticipatorily aligned before action (AAC) and how attribution stabilizes after events through investigative processes (RC). However, less attention has been paid to how computational infrastructures alter the tempo and breadth of stabilization across these stages. By conceptualizing AI as a representational accelerator rather than an autonomous decision-maker, PAC extends

accountability theory beyond attribution and compliance toward the temporal dynamics of responsibility production.

The FDA Warning Letter analysis establishes that responsibility convergence is not an artifact of computational mediation — it is a structural property of accountability documentation in regulated environments. Across 160 violation observations, documentary form produces uniform Firm-level attribution regardless of causal framing, evidentiary complexity, or violation type. This baseline finding reframes the theoretical puzzle: the question is not whether AI introduces convergence into organizational systems, but how it reshapes the tempo, breadth, and durability of convergence dynamics that already exist. PAC answers this question by specifying the mechanisms — representational compression, epistemic authority amplification, and temporal contraction — through which computational mediation intensifies pre-existing stabilization tendencies.

Within the broader responsibility lifecycle, responsibility is not a singular event but a staged organizational accomplishment. It is anticipatorily structured through procedural and evaluative expectations, embedded during execution through documentation architectures, stabilized retrospectively through investigative closure, and increasingly mediated through computational representation. PAC does not replace these stages; rather, it operates across them. By compressing interpretive possibilities, amplifying epistemic authority, and contracting deliberative time, computational mediation may intensify convergence dynamics that already exist within accountability infrastructures.

The central tension exposed by PAC is therefore not between automation and human judgment, but between speed and deliberation. As computational systems enhance administrative efficiency, they may simultaneously reduce the durability of ambiguity within organizational processes. Accountability systems differ not only in who they assign responsibility to, but in how long interpretive variance is permitted to endure before stabilization becomes institutionally binding. When interpretive space contracts too rapidly, distributed or structural explanations may lose durability before being meaningfully examined.

The scaffolded intervention findings extend this analysis toward design. The finding

that 15 of 20 participants produced single-cause explanations with confidence averaging just 1 out of 5 — before broadening their accounts with additional time or structured support — establishes that premature convergence is not a product of investigator certainty but of institutional pressure. The scaffold demonstrated that counter-convergence design can delay premature closure without automating conclusions or undermining compliance expectations, though replication at scale remains necessary. This finding has direct implications for how AI systems are deployed within investigative workflows: the design challenge is not improving the accuracy of computational outputs but preserving the conditions under which human judgment remains meaningfully comparative. AI deployed as a counter-convergence infrastructure — expanding interpretive space rather than compressing it, surfacing alternative causal accounts rather than foregrounding single attributions — represents a fundamentally different design orientation than AI deployed as an efficiency accelerator.

This perspective reframes AI not as an external disruptor of responsibility, but as an infrastructural layer that interacts with existing stabilization mechanisms. AI systems do not inherently produce new responsibility structures; instead, they reshape the timing and scope through which responsibility becomes administratively actionable. The central theoretical implication is therefore not technological determinism, but infrastructural acceleration. Responsibility convergence remains contingent on organizational ambiguity, closure pressure, and accountability design. Computational mediation may heighten these dynamics, particularly when representational outputs are granted authority and rapidly embedded in formal documentation.

Situating PAC within the responsibility program clarifies its relationship to adjacent frameworks. Anticipated accountability convergence (AAC) explains how expectations align before action through procedural and documentary infrastructures. Cognitive friction (CF) identifies how procedural texts embed localized judgment demands during execution — the specific textual configurations through which anticipatory expectations become operationalized at the sentence level of procedural language. Responsibility convergence (RC) explains how attribution stabilizes after events under investigative closure

pressure. PAC identifies how computational mediation may accelerate stabilization dynamics across these stages — and how deliberate counter-convergence design can resist that acceleration. Together, these frameworks trace responsibility structuring across the full temporal arc of regulated organizational work.

Accountability should therefore be understood not merely as a normative principle, but as an architecture of temporal structuring and evidentiary design. This integration suggests that accountability infrastructures must be analyzed not only in terms of fairness, transparency, or bias, but also in terms of temporal architecture. When computational systems compress interpretive space too rapidly, organizations may inadvertently reduce the durability of distributed explanations — producing accountability outcomes that are administratively efficient but causally impoverished.

Preserving interpretive plurality may therefore require deliberate design interventions, including mechanisms for contestability, temporal buffering, layered documentation that preserves alternative interpretations, and procedural conditions under which stabilized accounts can be reopened. Recognizing PAC as an infrastructural acceleration dynamic enables more precise analysis of how accountability architectures evolve under computational mediation — and how counter-convergence design can intervene in the temporal politics of responsibility stabilization.

8 Conclusion

This paper introduced Premature Accountability Convergence (PAC) as a conditional acceleration dynamic within contemporary accountability infrastructures. Rather than portraying AI as a determinant of responsibility, the framework conceptualized computational systems as representational mediators that may compress interpretive space and intensify stabilization dynamics within the responsibility lifecycle. PAC therefore reframes AI's role in organizations not as a source of new responsibility structures, but as an infrastructural force that alters the timing and scope of convergence.

The central contribution of this paper is not simply to observe that AI accelerates

accountability processes — a pattern increasingly noted in adjacent literatures — but to specify the structural mechanisms through which this acceleration operates, the empirical baseline against which it must be understood, and the design conditions under which it can be resisted. Document analysis of FDA Warning Letters establishes that responsibility convergence is a structural property of accountability documentation prior to any computational mediation: documentary form produces uniform attribution outcomes across causally heterogeneous violation descriptions, compressing interpretive diversity into administratively actionable accountability assignments regardless of evidentiary framing. This baseline clarifies the theoretical puzzle that PAC addresses — not the introduction of convergence, but its acceleration and the narrowing of the deliberative window within which alternative causal accounts remain viable.

The three mechanisms identified — representational compression, epistemic authority amplification, and temporal contraction of sensemaking — specify how computational mediation intensifies pre-existing stabilization tendencies within the responsibility lifecycle. These mechanisms do not operate independently; they interact and compound, producing acceleration dynamics that are structural rather than intentional. Organizations that deploy AI systems within investigative and evaluative workflows without attending to these mechanisms may inadvertently accelerate convergence toward actor-centered attribution at the expense of systemic causal reconstruction — reproducing the accountability patterns already observable in pre-computational regulatory documentation, but more rapidly and with diminished deliberative capacity.

The scaffolded intervention findings extend the framework toward design. Counter-convergence design — AI deployed to expand interpretive space rather than compress it, to surface alternative causal accounts rather than foreground single attributions — demonstrated that premature closure can be delayed without automating conclusions or undermining compliance expectations, though replication at scale remains necessary. This finding positions PAC not only as a diagnostic framework but as a foundation for accountability system design: the challenge is not improving the accuracy of computational outputs but preserving the temporal and representational conditions under which

human judgment remains meaningfully comparative.

Situated within a broader responsibility lifecycle, PAC specifies the AI-mediated stage at which pre-existing convergence dynamics are accelerated through computational representation — following anticipatory alignment (AAC), procedural embedding (CF), and investigative stabilization (RC). Together, these frameworks trace responsibility structuring across the full temporal arc of regulated organizational work, from the prospective alignment of accountability expectations through the procedural embedding of judgment demands, the investigative stabilization of attribution, and the representational acceleration of convergence through computational mediation. PAC thus completes the responsibility lifecycle by specifying how computational infrastructures reshape the temporal architecture of accountability in contemporary sociotechnical organizations — and by identifying counter-convergence design as the practical response to that reshaping.

Data Availability

FDA Warning Letters corpus: The 50 Warning Letters analyzed in this study are publicly available through the FDA's official database at `https://www.fda.gov/inspections-compliance-enforcement-and-criminal-investigations/compliance-actions-and warning-letters`. The violation-level coding dataset ($N = 160$) developed for this analysis, including attribution language classifications and violation cluster assignments, is publicly available at: `https://github.com/sorakim-lab/fda-warning-letters-pac-analysis`.

Pilot study and scaffold intervention data: Data from the two-study empirical program ($N = 20$ pilot; $N = 20$ scaffold intervention) consist of participant reasoning protocols collected in pharmaceutical investigation contexts. Due to participant confidentiality, these data are not publicly available but may be obtained from the author upon reasonable request at sorakim.lab@gmail.com.

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